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Topic : Hospital management system

Description

MediSync is a comprehensive Hospital Management System designed to streamline hospital operations by digitizing and automating core administrative, medical, and patient-centric tasks. The system allows administrators, doctors, nurses, and receptionists to work collaboratively, improving the efficiency, accuracy, and speed of operations such as patient admission, appointment scheduling, medical records management, billing, and inventory tracking.

This system is ideal for clinics, hospitals, or diagnostic centers aiming to reduce paperwork, avoid redundancy, and manage patient care more effectively while ensuring data integrity and security.

Tech Stack

Frontend

- **React.js** – Modern SPA development for hospital staff dashboard and patient portals
- **Tailwind CSS** – Utility-first styling framework for responsive design
- **Redux** – State management (for user sessions, appointments, etc.)

Backend

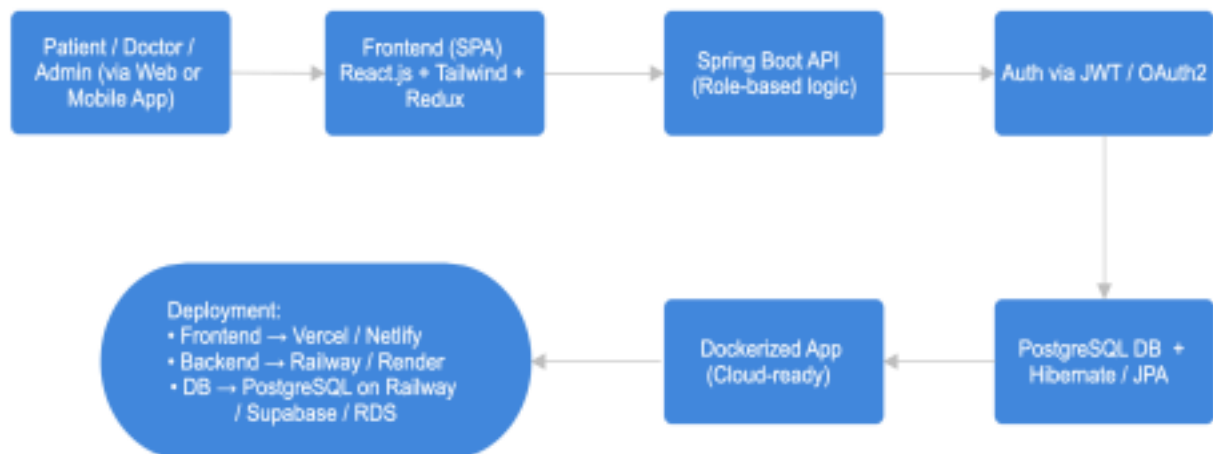
- **Spring Boot** – REST API for business logic and role-based workflows
- **JWT / OAuth2** – Secure authentication and role-based authorization
- **PostgreSQL** – Relational database to store hospital data
- **Hibernate / JPA** – ORM for database operations

DevOps / Hosting

- **Docker** – Containerization for local and cloud deployments
- **Render / Railway** – Backend deployment

- **Vercel / Netlify** – Frontend deployment

System Design



Core Functionalities

- Role-based dashboards (Admin, Doctor, Receptionist, Pharmacist)
- Patient registration and profile management
- Doctor scheduling and appointment management
- Electronic medical records (EMR)
- Prescription generation and record history
- Billing and payment tracking
- Inventory management (pharmacy/stock)
- Search and filter: Patients, doctors, departments

Authentication

- JWT-based login and session control

- Role-based access:
 - **Admin:** Full control over hospital records
 - **Doctor:** View appointments, medical records
 - **Receptionist:** Manage registrations and schedules
 - **Pharmacist:** Manage inventory

Future Scope

Short-Term Additions

- Email/SMS reminders for appointments
- Prescription PDF generation and email to patients
- Real-time notifications for critical stock levels
- Dashboard analytics for admin

Long-Term Features

- Telemedicine module (video consultation)
- AI-powered diagnosis suggestions based on symptoms
- Patient mobile app with appointment booking and medical history access
- Integration with wearable devices for live vitals tracking
- HL7/FHIR compliance for health data interoperability